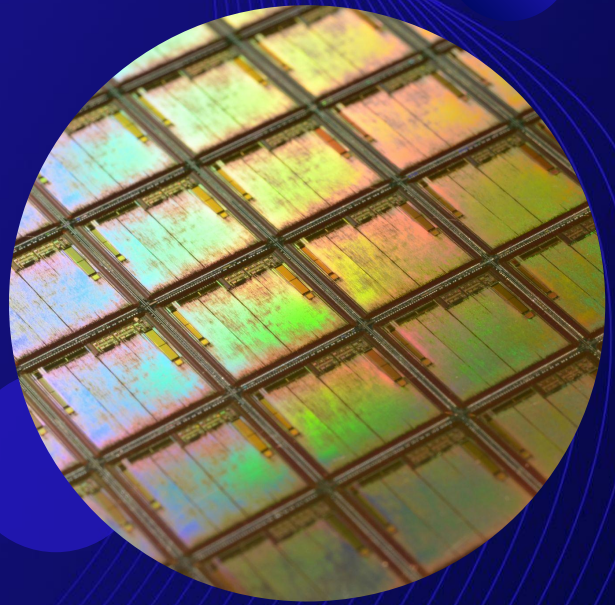
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The Fault in Our Cores

ThreadX, RISC-V, and Functional Safety



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Photo by Laura Ockel on Unsplash

About Me

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Developer, Architect, Product Manager...

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What Do They Have in Common?







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What Do They Have in Common?

Two Things

1. They're full of software (nowadays)
2. They can kill you (or others)

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What is actually a safe system?

- Safety means keeping the risk of harm at an **acceptable level**
- Functional safety depends on the system **responding correctly**
- The analysis includes hardware, software, people and environment
- Safety functions **prevent** hazardous events or **limit** their consequences
- Fault responses must act **before the system causes harm**



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How are hazards, risks, and safety integrity levels assessed?

- **Identify hazards** and the situations in which they could cause harm
- **Assess the severity** and likelihood of the consequences
- **Define safety functions** and the required risk reduction
- IEC 61508 assigns **SIL** 1–4 to safety functions
- ISO 26262 derives **ASIL**s from severity, exposure and controllability



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Why Reliable is not Necessarily Safe?

- Reliability concerns performing the **specified function over time**
- Safety concerns whether the system can cause **unacceptable harm**
- Correct software can faithfully implement unsafe requirements
    - A heater can reliably maintain a dangerously high temperature
    - An independent thermal cutoff provides a separate protective function



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ThreadX certification and safety artefacts

- Independent assessment under IEC 61508 and ISO 26262
- Certification covers defined releases and conditions
- Evidence includes safety manuals and verification records
- Reuse the artefacts to support the product's safety case
- RISC-V ports and integration need supporting evidence



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RISC-V architecture and SoC safety mechanisms

- Standardised PMP and Smepmp control CPU memory access (also, CHERI)
- ECC corrects covered memory errors and parity detects corruption
- Lockstep compares cores to detect execution mismatches
- Watchdogs and monitors supervise execution, clocks and power
- Fault coverage and responses depend on the SoC implementation



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Functional safety belongs to the whole system

- **Hazards** determine the safety functions and required integrity
- RTOS and SoC safety assumptions **must match the product**
- Fault detection and reaction must complete **within the safety deadline**
- Validate the **complete chain** from sensors to actuators
- The manufacturer demonstrates safety of the **complete product**



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The Big Picture

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12 Billion*

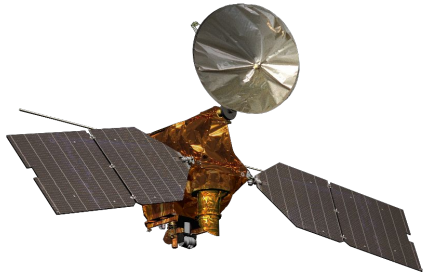
devices worldwide run ThreadX

* … and counting









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Eclipse ThreadX System Components

ThreadX
High-performance real-time operating system
(SIL 4, ASIL D, Med. Class C)

NetX Duo
IPv4/IPv6 stack with TLS/DTLS support
(SIL 4, ASIL D, Med. Class C)

FileX
Embedded FAT filesystem with optional fault tolerance

GUIX
Design environment and runtime for 2D user interfaces
(SIL 4, ASIL D, Med. Class C)

USBX
USB Stack providing host, device and on-the-go support
(SIL 4, ASIL D, Med. Class C)

TraceX
Graphical view of real-time events enabling analysis of system-level behavior

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Safety Certifications

SGS TÜV SAAR
FUNKTIONALE SICHERHEIT GEPRÜFT
FUNCTIONAL SAFETY APPROVED

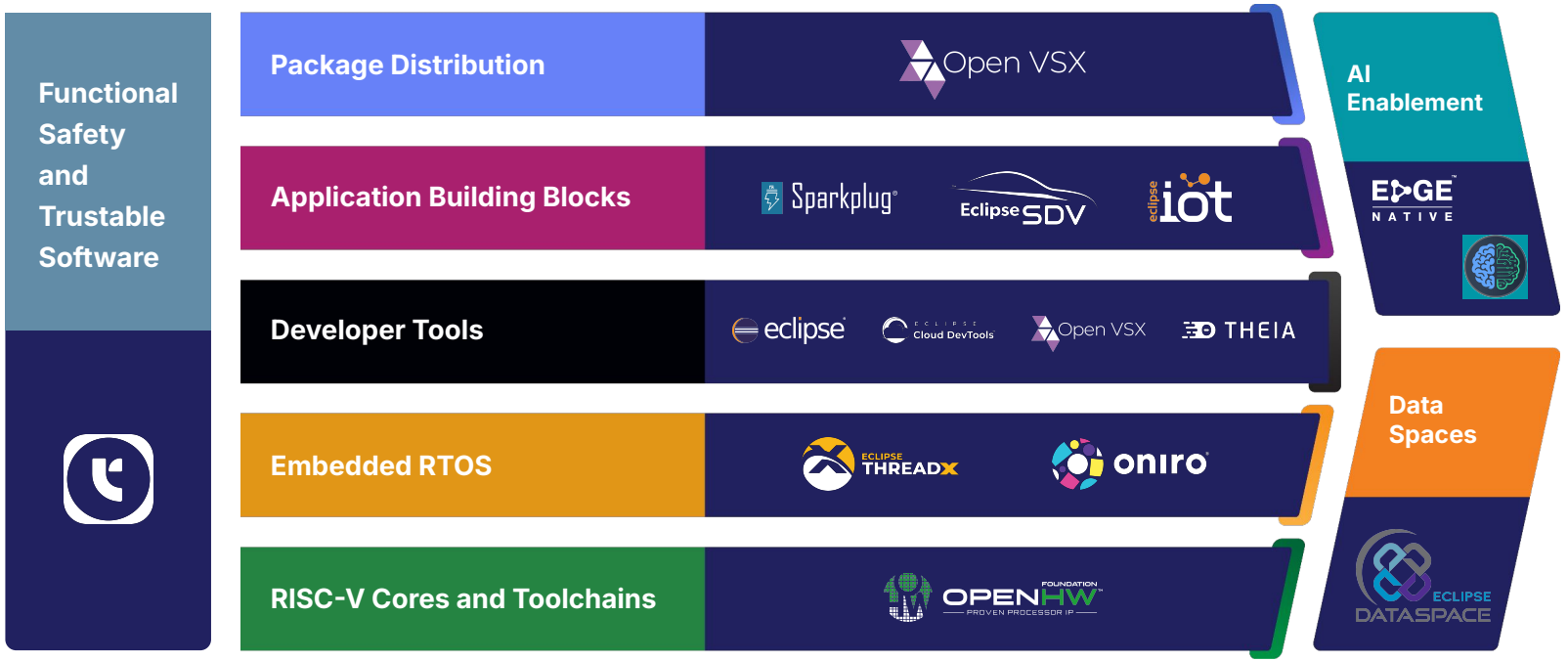
Certified by SGS-TÜV Saar for use in safety-critical systems, according to IEC-61508 SIL 4, IEC-62304 SW Safety Class C, ISO 26262 ASIL D and EN 50128

MISRA

Compliant with all "required" and "mandatory" rules of MISRA-C:2004 and MISRA C:2012

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Toward an Autonomous, Trustable and Scalable Stack



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